

→ probability rules :- (union)

① Addition rule :- The addition rule help us find the probability that atleast one of two event occurs.
ie The probability of A or B happening.

$$\rightarrow [P(A \cup B)]$$

→ General addition rule :- for any two events A & B.

$$[P(A \cup B) = P(A) + P(B) - P(A \cap B)]$$

→ $P(A \cup B)$ = probability of A & B occurring.

→ $P(A \cap B)$ = probability that both A & B occurs.

→ we subtract $P(A \cap B)$ to avoid double-counting that overlap.

Ex:- General Case [Not mutually exclusive]

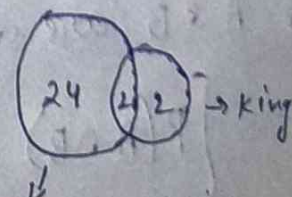
• What is probability of Drawing a red card or a King.

$$\rightarrow P(A \cup B) = P(A) + P(B)$$

$$= \frac{26}{52} + \frac{4}{52} - \frac{2}{52} \rightarrow \text{overlap}$$

$$= 0.5384 \times 100$$

$$= 53.84\%$$



Red Card.

Ex 2 :- Special Case :- Mutually Exclusive event.

If A & B are mutually exclusive (They cannot happen together). then

$$[P(A \cup B) = P(A) + P(B)]$$

② Multiplication rule :- The multiplication rule is used to find the probability of two events happening together. i.e. the probability of A & B occurring.

$$\rightarrow [P(A \cap B)]$$

Ex :- For Independent events :-

If A & B are independent then

$$[P(A \cap B) = P(A) \cdot P(B)]$$

use this when one event does not affect the other.

Ex 2 :- For Dependent events :-

If A & B are dependent then

$$[P(A \cap B) = P(A) \cdot P(B|A)]$$

use this when the probability of B depends on whether A has occurred.

→ Conditional probability :-

Conditional probability is the probability of an event A occurring given that another event B has already occurred.

→ It Denotes as $P(A|B) \rightarrow$ "The probability of A given B".

→ Formula :-
$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

→ Dependent events :- Dependent event means if the outcome of first event affects the outcome of second event.

→ There are 3 white balls & 2 red balls what is the probability of getting 2 white balls.

•
$$P(A \cap B) = P(A) * P(B|A)$$

•
$$P(A \cap B) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$$